Universiti Utara Malaysia	**STTHK2133 MODELING & SIMULATION** **A252 – (Group) Assignment #3** **Submission Date: 27 June 2026 via Learning Portal** Instructor: Assoc. Prof. Dr Azizi Ab Aziz

"As you start to walk on the way, the way appears." [Rumi]

Working in a pair is a great way to split the workload and brainstorm ideas! So, you are permitted to collaborate with another student on this assignment under the following conditions:

- Maximum Group Size: A maximum of **two (2)** persons per group is strictly enforced.
- Solo Option: You may choose to complete and submit this assignment individually if you prefer.
- Equal Contribution: Both partners are expected to contribute equally to the design, development, and final submission

Answer all questions (Part 1 and 2)

PART ONE (THEORETICAL SECTION)

1. Explain how micro-level agent rules can lead to macro-level system behavior. Why is this relationship important in Agent-Based Modeling theory?
2. Explain why ABMs are considered nonlinear systems, even if individual agent rules appear simple.
3. Design a simple ABM for modelling traffic congestion in a big city. Define:
    a) agents
    b) environment
    c) rules of interaction
4. What is meant by emergent behaviour in an agent-based model? Provide THREE (3) real-world example where emergence is observed.
5. Discuss why ABMs can become computationally expensive. What strategies are commonly used to improve scalability?
6. What is a digital twin? Explain how it differs from a simulation model.
7. Explain how real-time data integration changes the behaviour of a digital twin. What theoretical advantage does this provide over static simulation models?
8. Give TWO (2) real-world applications of digital twins in engineering, healthcare, or urban planning. Explain how they improve decision-making.?
9. In both ABM and digital twins, feedback loops are important. Explain what a feedback loop is and describe ONE (1) positive and negative feedback example.
10. What are the main limitations of agent-based models and digital twins in real-world applications? Discuss issues related to data, computation, and uncertainty.

PART TWO (PRACTICAL SECTION)

OPINION DYNAMICS IN A SOCIAL NETWORK

Social media platforms allow individuals to exchange opinions continuously. Through repeated interactions, opinions may converge toward consensus, become polarized, or fragment into competing groups. The Ministry of Education (MoE) plans to introduce AI-powered learning assistants in secondary schools and universities. The AI system can provide personalized tutoring, automated feedback, and learning analytics. Although experts claim the technology will improve educational outcomes, public opinion is divided. Supporters believe AI can improve learning efficiency, reduce teacher workload and personalize education. While the opponents worry about job displacement of teachers, student privacy, algorithmic biases and over-reliance on technology.

Public discussions occur primarily through social media platforms, online news portals, and community forums. Over time, citizens, influencers and experts interact, exchange opinions, consume media content, and react to misinformation campaigns. The government wants to understand:

- Will public opinion converge to support AI?
- Will society become polarized?
- Can misinformation significantly affect adoption?
- Which interventions can increase public acceptance?

Therefore, as a computational social scientist, you have been assigned to help the MoE to build an Agent-Based Model (ABM) to simulate how opinions evolve and to study whether society becomes supportive, divided (polarized), or neutral/mixed.

REQUIREMENTS (ROLES / AGENT) :

1. **Agent #1 (Citizens)** (Main Population). You must include 50–200 citizen agents, where each citizen has his/her own:
    - Opinion (O_i): range [-1, +1]; -1 = strongly against AI, 0 = neutral, +1 = strongly support AI
    - Trust (T_{ij}): trust in others (0–1)
    - Influence level: low (citizens have small influence)

   Role: Interact with other agents & update their opinion based on interactions.
2. **Agent #2 (Influencers)** (4–10 influencer agents) and this type of agent has high connectivity (many followers), strong influence on citizens and may have extreme or strong opinions.
   Role: Spread opinions quickly across the network, can shift public opinion significantly.
3. **Agent #3 (Influencers)** (2–5 expert agents) and this type of agent has high credibility (high trust from citizens), moderate influence (not viral, but respected) and usually balanced or evidence-based opinions.
   Role: Stabilize opinions, reduce extreme views and provide corrective or factual influence
4. **Network structure** – random network

Interaction Rules:

1. **Core opinion update** – when citizens interact with agent (education expert, influencer, other citizen opinion (averaging)), the opinion update is based on:
    - Direct influence from one connected agent (individual)- based on trust education expert effect and influencers strength
    - In addition, this rule models indirect social influence, where agents are affected by the average opinion of their neighbours, not just a single interaction.
    - Expected effect – gradual convergence
2. **Averaging Effect** - It pulls opinion toward "neighbour" average that represents social pressure:
    - It represents group discussion, social media feeds, comment threads and "what most people think" effect.
    - Expected effect - consensus pressure / echo chamber formation.
3. **Education expert effect** - stabilizes opinion via correction term, reduces extremism, and variance (pulls citizens towards expert's balanced view)
    - Expected effect – stability and moderation
4. **Influencer effect** - represents the stronger persuasive power of influencers in social media environments.
    - For example, higher frequency of post, likes and shares represent a high level of effect.
    - Expected effect - polarization or fast diffusion

INSTRUCTIONS (SIMULATION SETUP AND EXPERIMENTS)

Set up:

- Number of citizens: 50–200
- Influencers: 3–10
- Education experts: 1–5
- Time steps: 50–100
- Initial opinions: random between [-1, +1]

Simulation Tasks / Experiments for at least FOUR (4) scenarios:

- Scenario 1: Baseline Model (Observe whether opinions naturally converge or remain mixed)
    - Normal interaction between all agents
    - Moderate influence levels
- Scenario 2: Strong Influencer Impact (Analyze whether society becomes polarized or quickly aligned)
    - Increase influencer influence strength
- Scenario 3: Strong Expert Intervention (Check whether experts reduce polarization and stabilize opinions)
    - Increase trust in education experts
- Scenario 4: Low Trust Environment (Observe whether fragmentation or instability occurs)
    - Reduce trust across all agents

Your model must produce:

- Interactive system with functional GUIs
- 3D and 2D temporal plot of average opinion and individual opinion
- Histogram of final opinions
- Network visualization (optional)
- Short explanation of findings

Analysis (from simulation tasks):

1. For each scenario identify whether the system shows:
    a) Consensus ? Polarization ? Fragmentation ? Echo chambers ?
    b) Impact of each role
    c) System stability
2. Can a small number of influencers dominate public opinion? Explain using your simulation results.
3. Does increasing education expert influence always lead to consensus? Why or why not?
4. What happens when averaging effect is very strong? Is this realistic in real society?
5. Which factor is most important in preventing polarization: trust, experts, or network structure?
6. If you were a policy maker, how would you use education experts to stabilize public opinion?

SUBMISSION REQUIREMENTS:

- ABM source code (Python / Octave)
- Simulation results (graphs + tables)
- Short report (2–5 pages) including:
    - Model design
    - Equations
    - Experiments
    - Analysis and Discussion
    - Conclusion

Policy:

All grading of deliverables will be based on the standards indicated for each deliverable. Deliverables may not be turned in late, and no cheating! For this class, cheating will include plagiarism (using the writings of another without proper citation), copying of another (either current or past student's work), working with another on individually assigned work, or in any other way presenting as one's work that which is not entirely one's work. The occurrence of plagiarism will result in removal from the course with a failing grade.